
Program Name: Engineering

Branch: CE/IT

Course / Subject Code: BE02000041

Course / Subject Name: Fundamentals of AI

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Chapter 1: Introduction to Artificial Intelligence

3 Marks Questions

1. Define Artificial Intelligence and explain its significance in modern technology.
2. What is Artificial Narrow Intelligence (ANI)? Give an example.
3. What is the difference between Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI)?
4. Explain the concept of an intelligent agent in AI.
5. What are the characteristics of intelligent agents?
6. Define rationality in the context of AI agents.
7. Describe the nature of environments in AI.

4 Marks Questions

1. Explain the history and overview of Artificial Intelligence.
2. Discuss the concepts of production, agents, and environments in Artificial Intelligence.
3. What are the characteristics of intelligent agents? Explain with examples.
4. Differentiate between Artificial Narrow Intelligence, Artificial General Intelligence, and Artificial Super Intelligence

7 Marks Questions

1. What is Artificial Intelligence? Discuss the different types of AI, such as Artificial Narrow Intelligence, Artificial General Intelligence, and Artificial Super Intelligence, with examples.
 2. Explain the concepts of production, agents, and environments. Discuss the characteristics of intelligent agents and the concept of rationality in AI.
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Chapter 2: Knowledge Representation

3 Marks Questions

1. What is knowledge representation in Artificial Intelligence?
2. Define Natural Language Processing (NLP) and its importance in AI.
3. What is pattern recognition? Give an example.
4. What is an expert system in AI?

4 Marks Questions

1. Explain the concept of knowledge representation in AI.
2. Describe the role of Natural Language Processing (NLP) in Artificial Intelligence.
3. What is pattern recognition, and how is it used in AI applications?
4. Briefly explain the concept of Expert Systems in AI.

7 Marks Questions

1. Discuss the concept of knowledge representation in AI. How is it used to solve problems and make decisions in expert systems?

2. Explain the importance of Natural Language Processing in AI. Provide examples of NLP applications.
 3. Discuss the role of pattern recognition in AI. How does it help in decision-making and problem-solving?
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Chapter 3: Basics of Machine Learning

3 Marks Questions

1. What is learning from examples in machine learning?
2. What is supervised learning? Provide an example.
3. Define unsupervised learning. How does it differ from supervised learning?
4. Explain reinforcement learning and provide a simple example.
5. What is linear regression in machine learning?
6. What is logistic regression used for in machine learning?

4 Marks Questions

1. Explain the different forms of learning in machine learning: supervised, unsupervised, and reinforcement learning.
2. Describe simple machine learning models such as linear and logistic regression.
3. Explain the difference between supervised learning and unsupervised learning.
4. Discuss the concept of reinforcement learning with an example.

7 Marks Questions

1. Discuss the forms of learning in machine learning: supervised, unsupervised, and reinforcement. Provide examples and explain how each form works.
2. Explain simple machine learning models such as linear regression, logistic regression, and support vector machines (SVM). How are these models applied in real-world

problems?

Chapter 4: Deep Learning

3 Marks Questions

1. What is deep learning, and how does it differ from traditional machine learning?
2. Define a neural network and explain its basic structure.
3. What are the different types of deep learning models?
4. Provide an example of a deep learning application.

4 Marks Questions

1. Explain the concept of deep learning and how it is used in AI.
2. Describe the structure of a neural network and explain its working.
3. Discuss different types of deep learning models and their applications.

7 Marks Questions

1. What is deep learning? Explain the types of deep learning models and their applications in areas like computer vision and natural language processing.
 2. Describe neural networks in deep learning. Explain their structure and working, and provide examples of deep learning applications.
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Chapter 5: Modern Artificial Intelligence

3 Marks Questions

1. What are Large Language Models (LLMs)? Give an example.
2. Mention some common AI-based tools like ChatGPT, Gemini, or Bhashini.

3. What are the current issues in the field of Artificial Intelligence?

4 Marks Questions

1. Explain the concept of Large Language Models (LLMs) and their significance in AI.
2. Discuss the use cases of AI tools like ChatGPT, Gemini, Bhashini, and Krutrim.
3. What are the current issues and challenges in the development and application of Artificial Intelligence?

7 Marks Questions

1. Discuss the concept and applications of Large Language Models (LLMs). How do models like ChatGPT and Gemini impact various industries?
 2. Explain the current issues and future challenges in Artificial Intelligence. How can these challenges be addressed in the future development of AI technologies?
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